europass



Ayantika Gangopadhyay

Place of birth: Bhadreshwar , India | **Address:** A/3 Anandapuri Barrackpore, PO-NC Pukur , Kolkata- 700122, Barrackpore, India (Home)

ABOUT ME

Computer Science and Engineering undergraduate specialising in Cybersecurity and Forensics, with research experience in post-quantum cryptography, cryptographic implementation, and hardware security. Through research internships at IIT Roorkee, IIT Bhilai, and NIT Durgapur, I have explored post-quantum cryptographic optimisation and hardware-oriented security analysis. My research interests focus on hardware security, FPGA-based cryptographic implementation, RTL design, and efficient post-quantum cryptographic systems. I aspire to pursue advanced studies in cybersecurity and hardware security, contributing to the design and evaluation of secure, efficient, and resilient cryptographic architectures. Beyond engineering, I hold a Sangeet Visharad Final (5th Year) qualification in Rabindra Sangeet (Vocal) from Pracheen Kala Kendra, Chandigarh, with distinctions in both theory and practical examinations.

EDUCATION & TRAINING

20/07/2023 - Current - KOLKATA, INDIA

BACHELOR OF TECHNOLOGY (HONOURS) IN COMPUTER SCIENCE ENGINEERING WITH SPECIALISATION IS CYBERSECURITY AND FORENSICS- ADAMAS UNIVERSITY

Cryptography, Ethical Hacking, Penetration Testing, Information Security Audit & Monitoring, Computer Networks, Operating Systems, Database Management Systems, Computer Organisation and Architecture, Low Power VLSI Design, Switching Circuit and Logic Design, Mobile Communication

WORK EXPERIENCE

05/01/2026 - 05/02/2026 - DURGAPUR (ON-SITE), INDIA

RESEARCH INTERN, POST-QUANTUM CRYPTOGRAPHYNIT DURGAPUR

Implemented ML-DSA-87 (CRYSTALS-Dilithium, NIST FIPS 204) – key generation, signing, and verification – using lattice-based cryptography and polynomial ring arithmetic in Java

01/05/2026 - 30/06/2026 - KOLKATA (REMOTE), INDIA

RESEARCH INTERN, POST-QUANTUM CRYPTOGRAPHYIIT ROORKEE

Optimized the Number Theoretic Transform in ML-KEM (FIPS 203) with a fused Layer 5–6 butterfly kernel, achieving a 1.07× speedup on Apple M2 while maintaining 100% coefficient-level correctness.

04/06/2026 - 04/08/2026 - BHILAI (ON-SITE), INDIA

PROJECT INTERN, FPGA & CRYPTOGRAPHIC HARDWAREIIT BHILAI

Designed and verified digital hardware modules for AES-128 on the ChipWhisperer CW305 (Artix-7) FPGA, covering RTL design, synthesis, timing analysis, and bitstream generation; studied Differential Fault Analysis and fault-injection techniques

16/02/2026 - 28/02/2026 - RAJENDRA NAGAR (ON-SITE), INDIA

INDUSTRIAL TRAINEE, AI-AUGMENTED SYSTEM ANALYSISZENEEDA, NEW DELHI

Completed an industrial training program on AI-driven system analysis and problem-solving, gaining exposure to real-world industry workflows.

SKILLS

Languages
Java | Python | C++ | C | HTML | CSS | JavaScript | SQL

Security & Cryptography

Symmetric and Assymetric Cyrptography | Steam & Block Ciphers | Cryptanalysis | Differential Fault Analysis | Post quantum cryptography (ML-DSA, ML-KEM) | Penetration Testing | Web security | Ethical Hacking .

Hardware

Verilog HDL | xilinx vivado | ChipWhisperer | CMOS VLSI

ML/AI

scikit-Learn | Python (NumPy, Pandas, NLTK, Matplotlib, sklearn, TensorFlow) | CNNs | Adversarial Attack and Detection

Tools

IDEs: Eclipse, Visual Studio, PyCharm | Google Colab | Linux / MacOS terminal | Git / Git Hub | Wireshark | Cisco PacketTracer | Databases : MySQL , MongoDB

LANGUAGE SKILLS

Mother tongue(s): **BENGALI**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
HINDI	C2	C2	C2	C2	C2
ENGLISH	C1	C1	C1	C1	C1

PUBLICATIONS

Post-Quantum Authentication and Adversarial Robustness Evaluation for Secure CNN-Based Image Classification Model 2027

This paper introduces PQCML-Secure, a framework that protects AI image models from both pixel-tweaking adversarial attacks (FGSM) and file tampering during deployment. By combining robust adversarial retraining with a lightweight Dilithium-inspired post-quantum authentication system, the architecture successfully defends model integrity and maintains high classification accuracy. Tested on geometric and MNIST datasets, the system blocks hackers from fooling predictions or altering model files, offering a fast and highly secure pipeline for real-world AI deployment.n

Ayantika Gangopadhyay, Joy Chakraborty Springer Nature Cureus Journal,

Optimisation for the Number Theoretic Transform in ML-KEM on Apple M2 Architecture 2026

This paper optimizes the critical Number Theoretic Transform (NTT) operation in ML-KEM (FIPS 203) by fusing layers 5 and 6 into a single execution step. By keeping intermediate data entirely inside the CPU registers rather than bouncing it back and forth to main memory, this approach eliminates cache bottlenecks without altering the underlying cryptographic math. When benchmarked over 100,000 iterations on an Apple M2 processor, this register-level optimization achieves a localized performance boost of 7.27% while maintaining flawless arithmetic correctness.

Ayantika Gangopadhyay National Institute of Technology Goa 1st International Conference on Mathematics and AI : Modelling, Analysis and Computation (ICMAI-MAC). ICMAI-MAC 2026

Machine Learning-Based Prediction and Classification of Arsenic Contamination in Water 2026

This study develops a cost-effective machine learning framework to predict groundwater arsenic contamination in India using easily measurable water parameters like pH, turbidity, and temperature. By testing four algorithms on a realistic groundwater dataset, the research demonstrates that ensemble methods like Random Forest and Gradient Boosting deliver the highest predictive accuracy. Ultimately, this approach serves as a scalable early-warning system for rural public health monitoring and future IoT integration.e the description

Ayantika Gangopadhyay, Debasree Mitra, Riya Rabidas, Ankita Das Springer Nature Chapter 21 of Data Management, Analytics and Innovation: Proceedings of ICDMAI 2026, Volume 1 (ISBN 978-3-032-38256-6) corresponds to production code 688195_1_En within the Lecture Notes in Networks and Systems series (Volume 2162). . Data Management , Analytics and Innovation ICDMAI 2026, Volume 1, LNNS 2162 proceedings.

PROJECTS

12/08/2026 - CURRENT

Post-Quantum Secure and Crypto-Agile Digital Forensic Evidence Preservation Developed a post-quantum digital forensic evidence preservation framework integrating SHA-256, RSA-2048, ML-DSA-65, automated chain-of-custody tracking, and tamper detection for long-term evidence integrity.

Languages and framework- Python, ML-DSA

14/12/2025 - 20/04/2026

PQCML-Secure: Post-Quantum Authentication for CNNs Combined FGSM-based adversarial robustness evaluation with Dilithium-inspired digital signatures and SHA-256 hashing to detect tampered CNN models, achieving under 10ms signing/verification latency.

Languages and frameworks- Python, CNN, ML-DSA

02/01/2026 - 30/05/2026

Automated SQL Injection Detection Framework Built a web security testing tool for detecting SQL injection via crawling, form detection, and payload injection, with a Tkinter GUI and automated reporting; validated on vulnerable test applications.

Languages and Frameworks - Python, Requests, BeautifulSoup, Tkinter

09/09/2025 - 28/12/2025

ML-Based Prediction of Arsenic Contamination in Groundwater Compared Linear Regression, Random Forest, Gradient Boosting, and SVM to predict arsenic contamination from physicochemical parameters; Random Forest achieved the best performance

Languages and frameworks- Python, ML

HOBBIES AND INTERESTS

Bachelor of Music (B.Mus.) Equivalent – Sangeet Visharad (5th Year) Completed a 5-year comprehensive graduation program in Indian Classical Vocal Music from Pracheen Kala Kendra, Chandigarh (Session 2021–22).

Graduated with First Division honors, securing Distinction in both Music Theory and Practical examinations.